# CLINICAL CORRELATES

EMBRYOLOGY

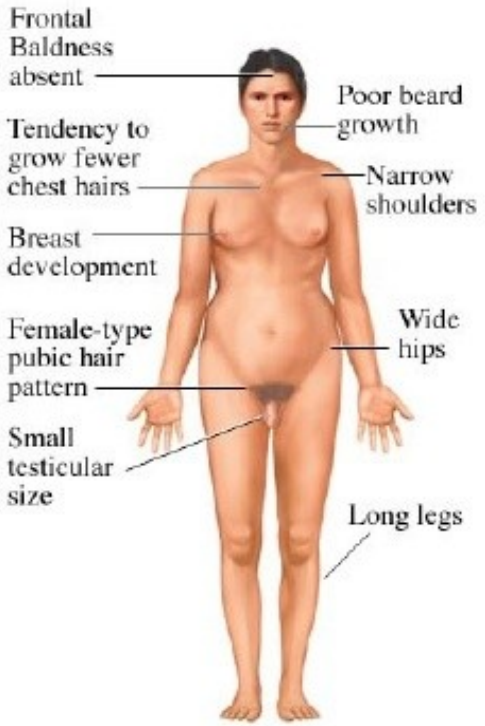
# DOWN SYNDROME



- **Down syndrome** (DS or DNS), also known as trisomy 21, is a genetic disorder caused by the presence of all or part of a third copy of chromosome 21. It is typically associated with physical growth delays, mild to moderate intellectual disability, and characteristic facial features

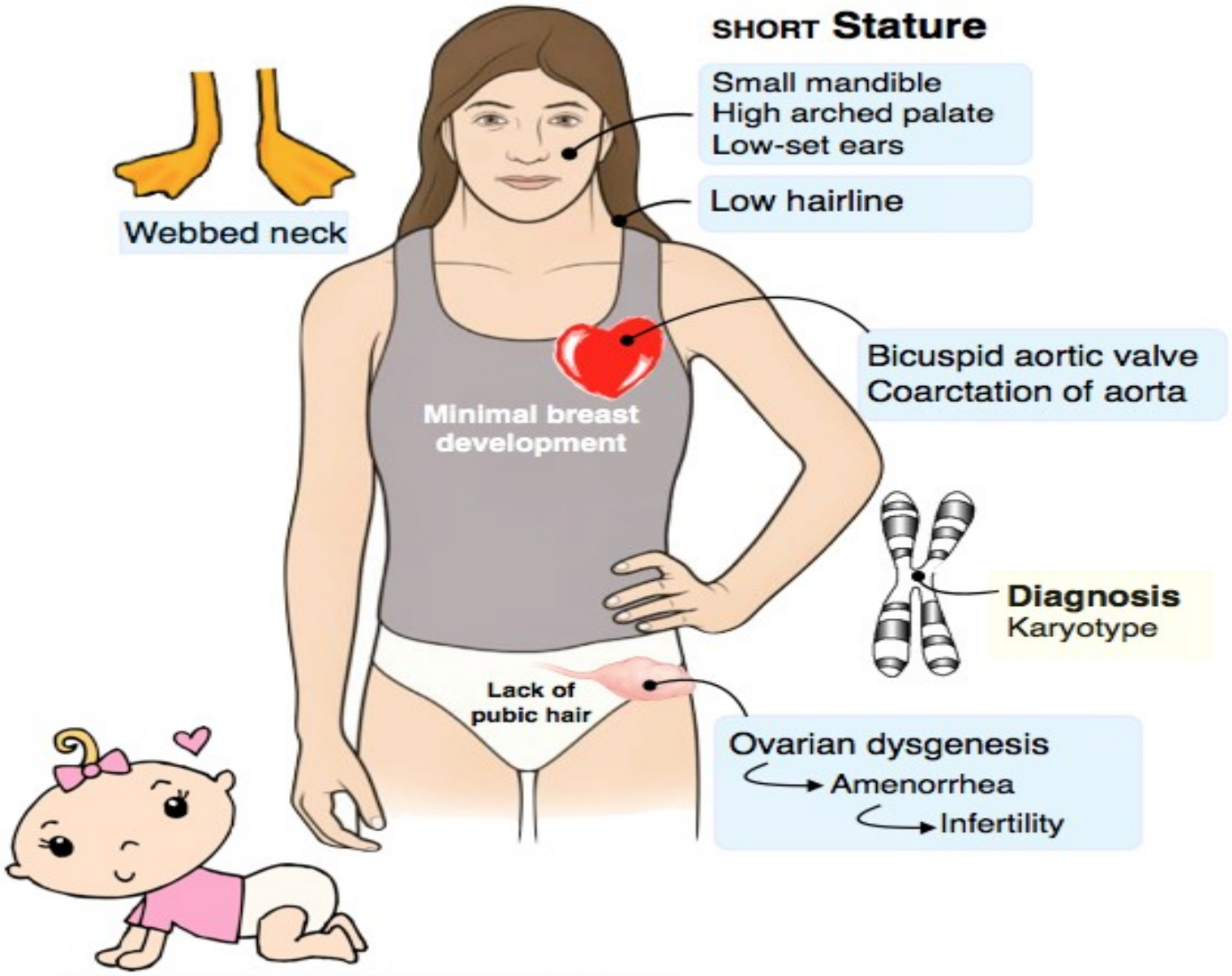
# Klinefelter syndrome

- Physical characteristics
  - *when present, may include*
    - sparse pubic, facial, and body hair
    - underdeveloped muscles
    - enlarged breasts (gynecomastia)
    - may also be taller than others in their family and have long legs, narrow shoulders, and wide hips




http://yalenewhavenhealth.org/library/healthguide/en-us/support/topic.asp?hwid=zm2596

# Turner Syndrome 45 XO




- Short 4th and 5th metacarpal bones
- Widely spaced nipples
- Lymphedema of hands/feet

# EDWARDS SYNDROME



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# PATAU SYNDROME




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**OLIGOZOOSPERMIA**

Terms oligospermia, **oligozoospermia**, and low sperm count refer to semen with a low concentration of sperm and is a common finding in male infertility. Often semen with a decreased sperm concentration may also show significant abnormalities in sperm morphology and motility

**AZOOSPERMIA**

**Azoospermia** is the medical condition of a man whose semen contains no sperm. It is associated with infertility, but many forms are amenable to medical treatment.

**Anovulation** is when the ovaries do not release an oocyte during a menstrual cycle. Therefore, ovulation does not take place. However, a woman who does not ovulate at each menstrual cycle is not necessarily going through menopause. Chronic anovulation is a common cause of infertility.

***Polymenorrhea*** is the medical term for cycles with intervals of 21 days or fewer.
*Irregular menstruation* is where there is variation in menstrual cycle length of more than approximately eight days for a woman.
***metrorrhagia*** is often used for irregular menstruation that occurs between the expected menstrual periods.
***Oligomenorrhea*** is the medical term for infrequent, often light menstrual periods (intervals exceeding 35 days).
***Amenorrhea*** is the absence of a menstrual period in a woman of reproductive age. Physiologic states of amenorrhoea are during pregnancy and lactation (breastfeeding). Outside of the reproductive years there is absence of menses during childhood and after menopause.

Abnormal uterine bleeding is a general category that includes any bleeding from menstrual or nonmenstrual causes.

Hypomenorrhea is abnormally light menstrual periods.

Menorrhagia (meno = month,[2] rrhagia = excessive flow/discharge) is an abnormally heavy and prolonged menstrual period.

Metrorrhagia is bleeding at irregular times, especially outside the expected intervals of the menstrual cycle. If there is excessive menstrual and uterine bleeding other than that caused by menstruation,

menometrorrhagia (meno = prolonged, metro = uterine, rrhagia = excessive flow/discharge) may be diagnosed. Causes may be due to abnormal blood clotting, disruption of normal hormonal regulation of periods or disorders of the endometrial lining of the uterus. Depending upon the cause, it may be associated with abnormally painful periods.

**Dysmenorrhea** (or dysmenorrhoea), cramps or painful menstruation, involves menstrual periods that are accompanied by either sharp, intermittent pain or dull, aching pain, usually in the pelvis or lower abdomen.

**Placenta Previa** is a condition where the **placenta** lies low in the uterus and partially or completely covers the cervix. The**placenta** may separate from the uterine wall as the cervix begins to dilate (open) during labor.

**Ectopic pregnancy** is a complication of **pregnancy** in which the embryo attaches outside the uterus. Signs and symptoms classically include abdominal pain and vaginal bleeding. Fewer than 50 percent of affected women have both of these symptoms. The pain may be described as sharp, dull, or crampy.

**Choriocarcinoma** is a fast-growing cancer that occurs in a woman's uterus (womb). The abnormal cells start in the tissue that would normally become the placenta. This is the organ that develops during pregnancy to feed the fetus. **Choriocarcinoma** is a type of gestational trophoblastic disease.

**Invasive mole** is unlike **choriocarcinoma**, the latter is without the presence of chorionic villi. ... The clinical presentation of an **invasive mole** includes vaginal bleeding, an enlarged uterus and high urinary or serum bHCG level, typically after the evacuation of a **molar pregnancy**.

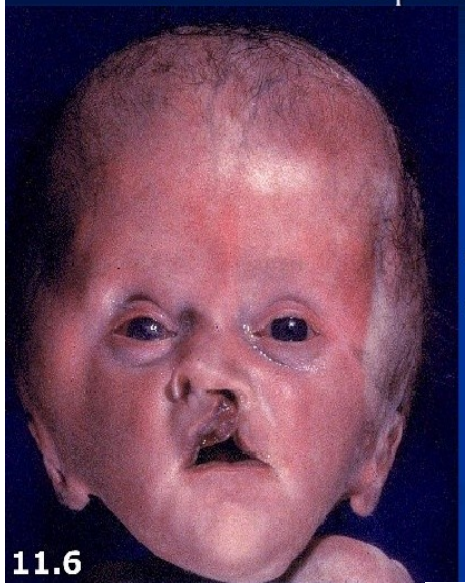
A **hydatidiform mole** is growth of an abnormal fertilized egg or an overgrowth of tissue from the placenta. Women appear to be pregnant, but the uterus enlarges much more rapidly than in a normal pregnancy.

**Holoprosencephaly** (HPE) is a cephalic disorder in which the prosencephalon (the forebrain of the embryo) fails to develop into two hemispheres. Normally, the forebrain is formed and the face begins to develop in the fifth and sixth weeks of human pregnancy.

**Caudal regression syndrome**, or sacral **agenesis** (or hypoplasia of the sacrum), is a congenital disorder in which there is abnormal fetal development of the lower spine—the **caudal** partition of the spine. It occurs at a rate of approximately one per 25,000 live births.

**Sacrococcygeal teratoma** (SCT) is an unusual tumor that, in the newborn, is located at the base of the tailbone (coccyx). This birth defect is more common in female than in male babies. Although the tumors can grow very large, they are usually not malignant (that is, cancerous).

# What is Holoprosencephaly?



11.6

The Prosencephalon is divided into the two hemispheres:

- Left Hemisphere
- Right Hemisphere

















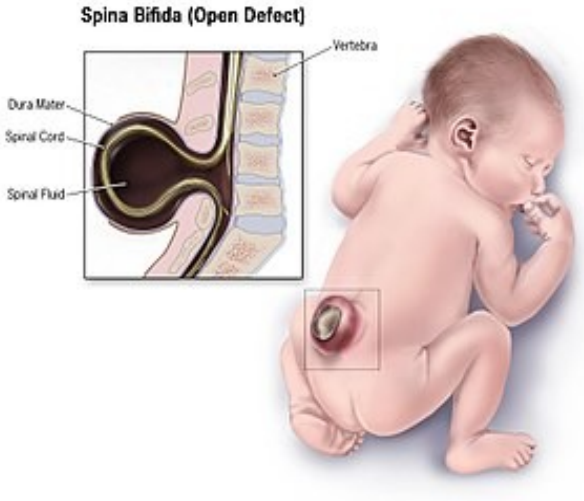

**Kartagener syndrome** is a type of primary ciliary dyskinesia that is also characterized by situs inversus totalis (mirror-image reversal of internal organs ).

**Anencephaly** is the absence of a major portion of the brain, skull, and scalp that occurs during embryonic development. It is a cephalic disorder that results from a neural tube defect that occurs when the rostral (head) end of the neural tube fails to close, usually between the 23rd and 26th day following conception.

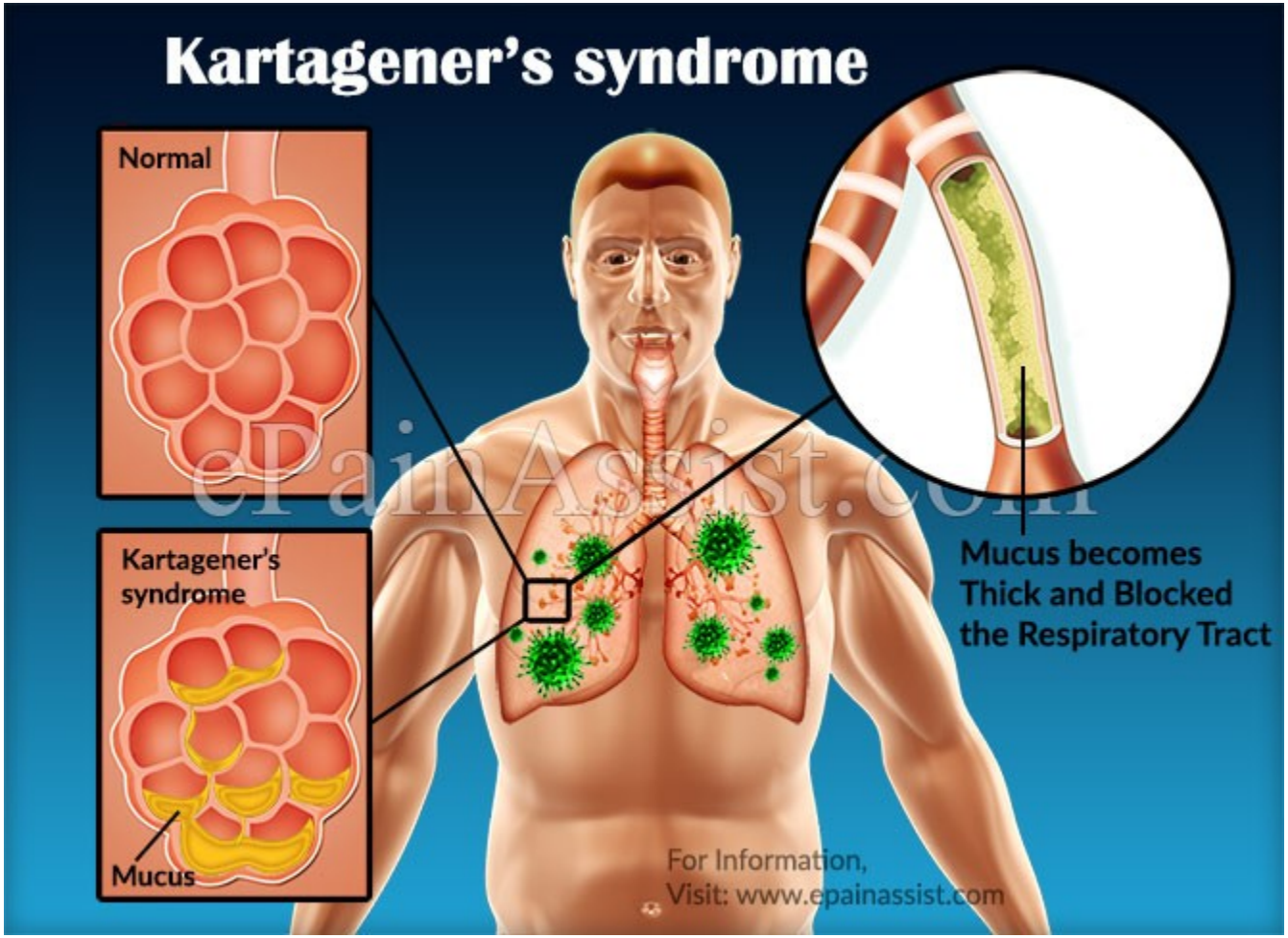
**Spina bifida** is a birth defect that occurs when the spine and**spinal** cord don't form properly. It falls under the broader category of neural tube defects. The neural tube is the embryonic structure that eventually develops into the baby's brain and**spinal** cord and the tissues that enclose them.

**Capillary hemangioma** is a vascular anomaly.**Capillary hemangiomas** occur 5 times more often in female infants than in males, and mostly in Caucasian populations. Additionally, low birthweight infants have a 26% chance of developing a **hemangioma**. It is the most common tumor of orbit and periorbital areas in childhood.




Spina Bifida (Open Defect)
Vertebra
Dura Mater
Spinal Cord
Spinal Fluid




Kartagener's syndrome
Normal
Kartagener's syndrome
Mucus
Mucus becomes Thick and Blocked the Respiratory Tract
ePainAssist.com
For Information,
Visit: www.epainassist.com